

Section A

Answer **all** the questions in this section

- 1 Fig 1.1 shows the relationship between certain organelles in a phagocytic cell.

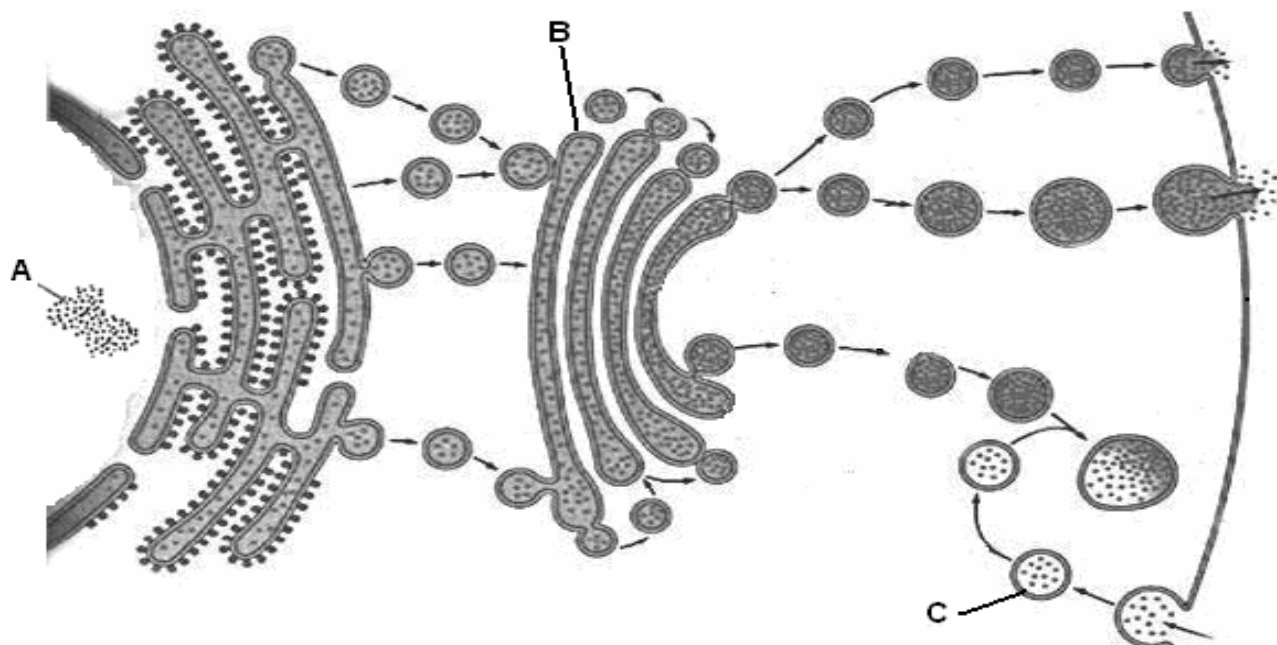


Fig 1.1

- (a) Identify organelles **B** and **C**. [2]

- **B Golgi body / Golgi apparatus**
- **C phagocytic/food/endocytic vacuole/vesicle**

- (b) State one function **B** may have in the cell. [1]

- **Further modification of proteins;;**
- **Sorts, and packages proteins into secretory vesicles for transport out of cell/ reference to packaging;;**
- **lysosome formation;;**
- **Synthesis of secretory polysaccharides/vesicles for exocytosis;;**

Lysosomes are membrane-bound vesicles that contain hydrolytic enzymes. One such enzyme is lysozyme as shown in Fig 1.2. Lysozyme is made up of 129 amino acids. Two amino acids present in the active site, Glu-35 and Asp-52, are vital for the catalytic action of lysozyme.

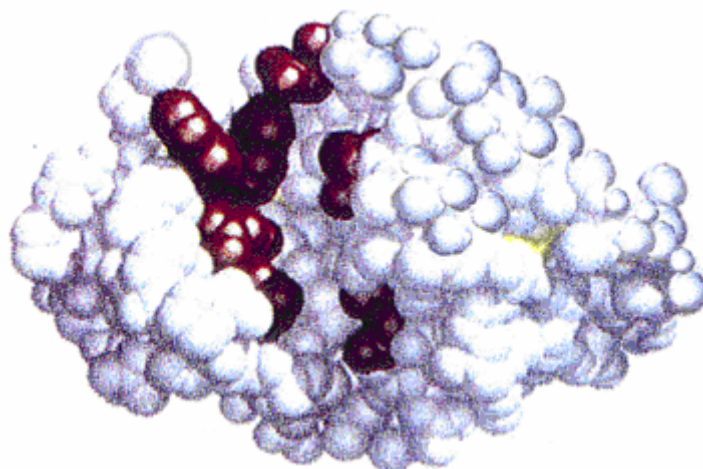


Fig 1.2

(c) Label with the letter **X** on Fig 1.1, a structure where the lysozymes are synthesized. [1]

(d) What is the part played by **A** in the synthesis of lysozymes? [1]

- RNA synthesis; ribosome assembly;

(e) Explain how these two amino acids at positions 35 and 52 in the amino acid sequence are brought close together to form the active site. [3]

- **Lysozyme contains a sequence of amino acids held together by peptide bonds (in a polypeptide chain);**
- **containing information for protein's folding into specific shape;**
- **(the two amino acids are brought close together) when the polypeptide chain fold into (geometrically regular) secondary structure;**
- **α -helix and β pleated sheet;**
- **as a result of hydrogen bonding between the amino acids;**
- **(From the figure), lysozyme is a globular protein;**
- **The polypeptide chain will undergo further folding to form the tertiary structure;**
- **maintained by the ionic bonds, hydrogen bonds, disulphide bonds and hydrophobic interactions;; (2 bonds = ;)**

[Total: 8]

- 2 Meat is the muscle of an animal. When animals are slaughtered, biochemical changes occur in the muscles, resulting in the hardening of meat. There are several ways to make the meat tender again, and these usually involve the partial breaking down of the proteins that make up the muscle tissue.

One way is to add the protein-digesting enzyme, papain, just before freezing the meat. As the meat is thawed and cooked, this enzyme digests the protein.

- (a) Suggest how the protein may be held in the active site of a papain while the reaction occurs. [2]

- **Spatial fit;**
- **substrate with shape complementary to the shape of the active site of the enzyme to bind to the enzyme's active site;;**
- **Substrate and enzyme are held at correct angle and orientation to ensure successful collision to form enzyme-substrate complex;;**

OR

- **Chemical fit;**
- **enzyme interacts with the substrate via hydrophobic interactions, hydrogen bonds and ionic bonds;;**
- **These bonds help bind the substrate to the enzyme, forming a enzyme-substrate complex at the active site;;**
- **The types of bonds formed depends on the nature of the R group of the amino acids that made up enzyme's active site;;**

- (b) Use your knowledge of enzymes to explain:

- (i) Why the rate at which protein is digested increases as the meat warms up in the early stages of cooking. [2]

- **In early stages of cooking, heating increases the kinetic energy of both papain and protein molecules in the muscle;;**
- **rate of successful collisions between enzyme and protein molecules increase;**
- **Resulting in formation of more enzyme-substrate complexes per unit time ;**

- (ii) Why protein digestion stops once the meat has been heated to 90°C. [3]

- **At 90°C, large amount of heat present resulting in denaturation of papain;**
- **3° bonds maintaining the 3D conformation of enzyme are broken;**
- **Causing loss of active site / change in shape of active site;**
- **Leading to loss of enzyme / catalytic activity / function of papain ;**
- **High heat also cause denaturation of muscle protein/breaks bonds maintaining 3D shape of muscle protein;**
- **Causing loss of specific shape of protein;**
- **Substrate can no longer fit into active site of enzyme;**
- **Hence no protein digestion could occur even if papain molecules are not denatured;**

[Total: 7]

- 3 (a) In the space below, draw a labelled diagram of a section through a mitochondrion. [2]

2 labels – 1 mark

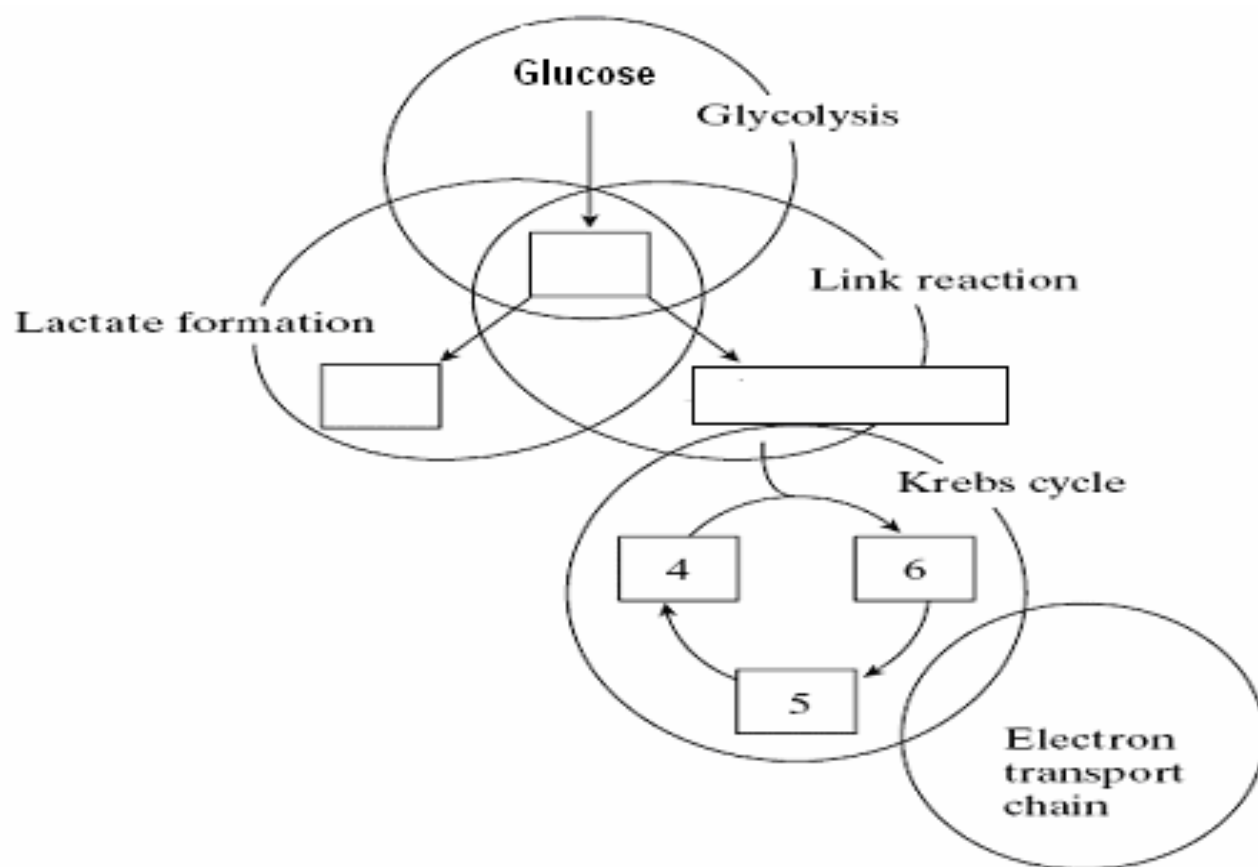
Correct structure – 1 mark

- (b) It has been suggested that mitochondria have evolved from free-living prokaryotic cells that were engulfed by a larger eukaryotic cells.

Compare the similarities between the mitochondrial and prokaryotic genome. [2]

- **Single circular, double stranded DNA molecule;;**
- **No histones associated / associated with nucleoid associated proteins;;**
- **Semi-autonomous reproduction, ie. reproduce with little reference to nuclear DNA/able to reproduce independent of nuclear DNA;;**
- **Mitochondrial genome is similar in size to prokaryotic genome size / contains about 35-40 genes/15,000-20,000 bases which is similar to that of prokaryotic genome size;;**

The diagram below summarizes the five main stages in respiration in an animal cell. The number of carbon atoms in various molecules or ions is shown in the boxes.



(c) Complete the diagram by filling in the **three empty boxes** with the names of the molecules. [2]

- **pyruvate, lactate, acetyl-coA**

2 correct – 1 mark

3 correct – 2 marks

(d) Name **two** stages shown in the diagram in which carbon dioxide is produced. [1]

- **Link reaction;**
- **Krebs cycle;**

(e) Red blood cells do not contain mitochondria but they use ATP. By what process do red blood cells produce ATP? Suggest a reason for your answer. [1]

- **Anaerobic respiration/ glycolysis/ lactate fermentation;**
- **Only cytosol present;**

(f) Describe the main stages of glycolysis. [4]

- Glucose is phosphorylated by 1 molecule of ATP to form Glucose-6-phosphate;
- Glucose-6-phosphate is isomerised to form Fructose-6-phosphate;
- Fructose-6-phosphate is phosphorylated by 1 molecule of ATP to form Fructose-1,6-biphosphate;
- Fructose-1,6-biphosphate is split to form Triose phosphate;
- 2 molecules of Triose phosphate are phosphorylated by 2 molecules of inorganic phosphate and oxidised by two molecules of NAD^+ to form 2 molecules of Glycerate biphosphate;
- 2 molecules of Glycerate biphosphate are converted to form 2 molecules of Glycerate phosphate;
- 2 molecules of ATP are formed by substrate level phosphorylation (SLP);
- 2 molecules of Glycerate phosphate are converted to form 2 molecules of pyruvate ;
- 2 molecules of ATP are formed by substrate level phosphorylation (SLP);

Triose phosphate= Glyceraldehyde-3-phosphate
Glycerate biphosphate = 1,3-diphosphoglycerate
Glycerate phosphate = 3-phosphoglycerate

[Total: 12]

4 Fig. 4.1 shows part of a motor neuron.

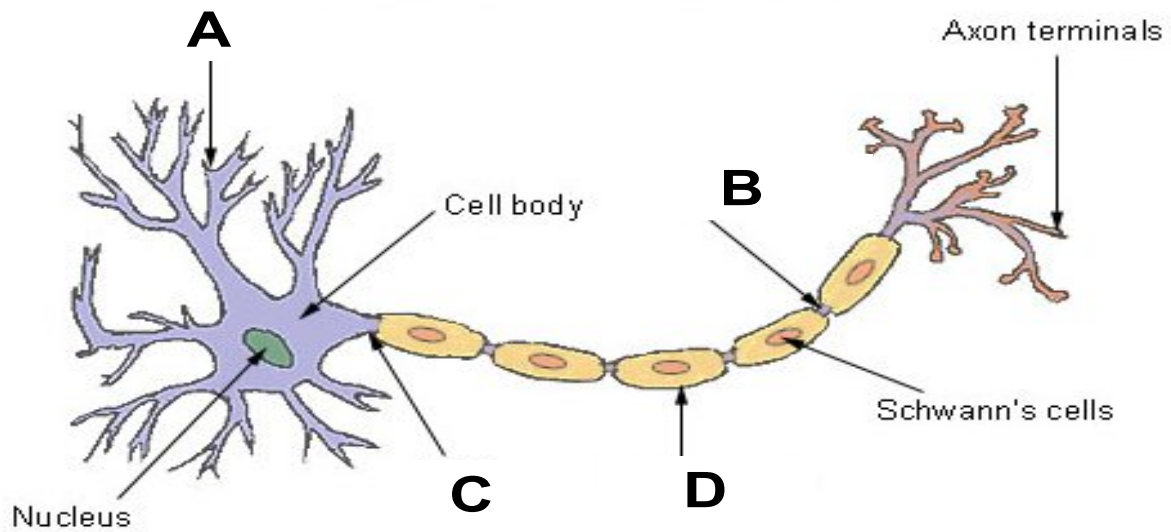


Fig 4.1

(a) Name the regions **A** and **B**. [2]

- **A = Dendrite;**
- **B = Node of Ranvier;**

(b) Explain the significance of structures **C** and **D** in nerve impulse transmission. [4]

- **C - Axon;**
- **It transmits the nerve impulse;**
- **From the cell body of the motor neurone;**
- **to the synaptic terminals;**
- **Important for propagating the electrical impulse / nerve impulse / action potential;**
- **D - Myelin sheath;**
- **Acts as a form of insulation;**
- **Electrical impulse / nerve impulse / action potentials are propagated along a myelinated axon through the nodes of Ranvier;**
- **by saltatory conduction;**
- **Increasing the speed of nerve impulse propagation;**

Fig. 4.2 shows a synapse between two neurons.

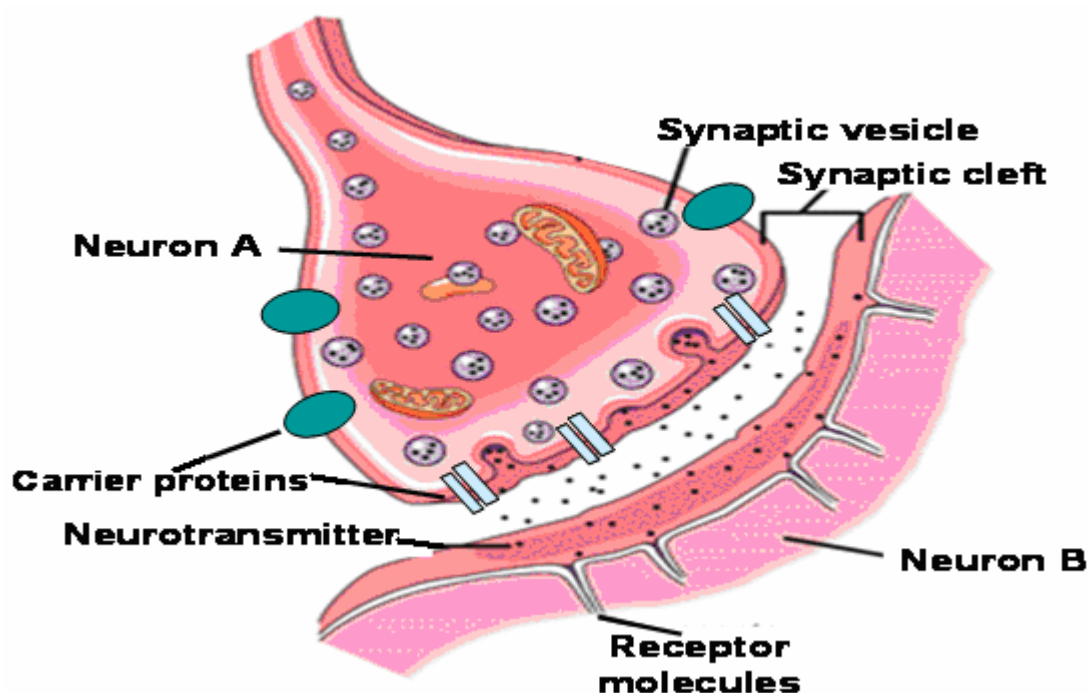


Fig 4.2

(c) Describe the process which leads to the release of the neurotransmitter from neuron A into the synaptic cleft. [3]

- **Arrival of the electrical impulse / nerve impulse / action potential at the synaptic knob;**
- **(depolarization) causes the opening of gated calcium ion channels;**
- **Influx of calcium ions by diffusion;**
- **causing synaptic vesicles containing neurotransmitter to move to the pre-synaptic membrane;**
- **fuse with the membrane;**
- **releasing its neurotransmitter via exocytosis;**

(d) Suggest the function of carrier proteins in neuron A. [1]

- **Influx of calcium ions through gated calcium ion channels;;**
- **Reabsorption of neurotransmitter;;**

(e) Explain how the neurotransmitter causes a nerve impulse to be initiated in neuron B.
[3]

- Neurotransmitter binds to specific receptor sites/molecules on the postsynaptic membrane;
- Na⁺ Gated channels open;
- sodium ions diffuse down concentration gradient/region of higher concentration to a region of lower concentration;
- Depolarising the post synaptic membrane;
- an excitatory post-synaptic potential (EPSP) is formed;
- If the depolarization/ summation of EPSP reaches/exceed the threshold potential, an nerve impulse is generated down neuron B

[Total: 13]

- 5 A farmer crossed pure breeding long grained rice with brown husk with pure breeding short grained rice with white husk. All F_1 offspring showed long grained rice with brown husk. A test cross was made between the F_1 plant and a pure-bred plant with short grained rice with white husk. The phenotypes of 200 offsprings of the test cross was recorded and shown in Table 5.1

Table 5.1

Long grained rice with brown husk	Short grained rice with brown husk	Long grained rice with white husk	Short grained rice with white husk
90	8	6	96

(a) Using the following symbols, draw a genetic diagram to explain the results of the test cross. [4]

L – allele for long grained rice
l – allele for short grained rice
B – allele for brown husk
b – allele for white husk

P1 phenotype	long grained, brown husk		x	short grained, white husk	
P1 genotype	$\frac{LL}{ll} \frac{BB}{bb}$			$\frac{ll}{ll} \frac{bb}{BB}$	
Gametes	$\frac{LB}{LB}$	$\frac{Lb}{Lb}$ $\frac{lB}{lB}$ $\frac{lb}{lb}$		$\frac{lB}{lB}$	$\frac{lb}{lb}$
	$\frac{LB}{LB}$	$\frac{Lb}{Lb}$		$\frac{lB}{lB}$	$\frac{lb}{lb}$
$\frac{lB}{lB}$	$\frac{LB}{lB}$	$\frac{Lb}{lB}$		$\frac{lB}{lB}$	$\frac{lb}{lB}$
	$\frac{LB}{lb}$	$\frac{Lb}{lb}$		$\frac{lB}{lb}$	$\frac{lb}{lb}$
	Long grain, brown husk	Long grain, white husk		Short grain, brown husk	Short grain, white husk
Phenotypic no	90	6		8	96
Phenotypic ratio	15	1		1	16

- (b) Carry out a chi-squared (χ^2) test to determine if the results of the test cross depart significantly from the expected ratio. [4]

Table 5.2

degrees of freedom	Distribution of X^2				
	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$v = c - 1$$

keys to symbols

Σ - 'sum of...'

O - observed value

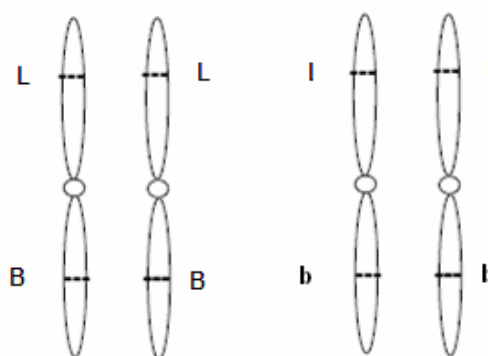
E - expected value

v - degrees of freedom

c - number of classes

- $\chi^2 = 148.3$;
- Working;
- Deg of freedom = 3;
- (Calculated) X^2 value of 148.3 is more than;
- Critical X^2 value of 7.82 at $P = 0.05$;
- reject H_0 ;
- results of the test cross depart significantly from the expected ratio;
- the difference between the expected ratio and the observed ratio is significant;
- the deviation/difference is not due to chance but due to other factors;

- (c) Using your answer in (b), illustrate the distribution of the two gene loci for grain length and husk colour of the two pure breeding parent plants. [1]



(d) Explain the deviation of the observed test cross ratio from the expected test cross ratio. [3]

- **Expected dihybrid test cross ratio is 1:1:1:1 while observed test cross results do not have a definite ratio/15:1:1:16;;**
- **indicates that the 2 genes for grain length and colour of husk are linked;;**
- **As such, there is crossing over between the 2 linked genes (in only some of the cells);;**
- **Thus the observed offspring of test cross are majority of parental combinations and minority of recombinants;;**

There are three types of coat colour observed in the Norwegian cattle - albino, brown and black. Brown and black pigments are produced by the action of the enzyme tyrosinase.

Tyrosinase is coded for by the **E** locus, which has two alleles, **E** and **e**. No enzyme is produced when the recessive allele **e** is present. A low level of activity of the enzyme leads to the production of brown pigment, whilst a high activity allows black pigment production.

The dominant allele of a second gene, **A** codes for a peptide which binds to and inhibits tyrosinase activity. The other allele, **a**, results in no peptide production.

(e) State the coat colours of Norwegian cattles with the following genotypes: [2]

Genotypes	Coat colour
Aaee	Albino
AaEe	Albino
aaEE	Black

2 correct – 1 mark

3 correct – 2 marks

(f) Explain how cattles with brown coats arise. [2]

- **Possible genotype: aaEe;**
- **Cattles which are homozygous recessive for the gene which produce peptide that binds to and inhibits tyrosinase;**
- **Thus enzymatic activity is not inhibited;**
- **Cattles must be heterozygous for the gene which codes for tyrosinase;**
- **This will result in a low level of tyrosinase activity, leading to production of brown pigment;**

[Total: 16]

- 6 The mis-expression of critical genes can disrupt normal cellular processes and cause unregulated growth, leading to cancer.

The Ras protein, encoded by the *ras* gene, is a G protein that relays a signal from a growth factor receptor on the plasma membrane to a cascade of protein kinases. The conversion of the *ras* proto-oncogene to *ras* oncogene is a frequent occurrence found in colorectal carcinomas.

Only a small fraction of all cancers involve an inherited predisposition. Most develop sporadically in the population, without any obvious familial pattern. Nevertheless, in 5-10% of breast cancer cases, there is evidence of a strong inherited predisposition. *BRCA1* and *BRCA2* are two examples of tumour-suppressor genes. Mutations in either gene increase the risk of developing breast cancer and ovarian cancer.

- (a) State what you understand by the terms,

(i) Proto-oncogene [1]

- **Gene encoding for proteins; that are involved in normal cell division and cell growth;**

(ii) Tumour suppressor gene [1]

- **Gene encoding for proteins; that normally function in preventing inappropriate cell growth;**

- (b) Describe **one** genetic change that could convert normal proto-oncogenes to abnormal oncogenes, resulting in gain-of-function mutations. [2]

- **Amplification;;**
- **Amplification of the proto-oncogene;**
- **Abnormal increase in the copy number of the proto-oncogene;**

OR

- **Point mutation;;**
- **Point mutation in the proto-oncogene or control element;**
- **Increase in its expression;**
- **Oncogenic protein is more active or more resistant to degradation than the normal protein;**

OR

- **Translocation / Transposition;;**
- **Translocation of the proto-oncogene to new locus under new control (eg. active promoter/enhancer) OR translocation of a more active promoter near a proto-oncogene;**
- **Transcription may increase, making it an oncogene;**

OR

- **Viral integration;;**
- **Viral integration into the chromosome;**
- **Result in mutation that increase expression of nearby proto-oncogenes;**

- (c) Research showed that only 5-10% of breast cancer cases have evidence of a strong inherited predisposition. Suggest why most breast cancers are considered non-hereditary? [1]

- **Most breast cancers are associated with somatic mutations / not germ-line mutation;;**

Two drugs, Tamoxifen and Bevacizumab are used to treat breast cancer depending on the stage of cancer progression.

Tamoxifen is a type of drug called selective estrogen receptor modulator (SERM). SERMs act by blocking any estrogen present in the body from attaching to the estrogen receptor on the cancer cells, eventually slowing or preventing the growth of tumors and killing tumor cells.

Bevacizumab, a targeted drug, attacks specific abnormalities within cancer cells. It is a drug designed to stop the signals cancer cells use to attract new blood vessels.

(d) Suggest what is the normal function of estrogen? Explain your answer. [2]

- **SERMs blocks estrogen from binding;**
- **And therefore preventing the growth of tumours;**
- **Suggesting that the normal function of estrogen is to stimulate cell growth;;**

Fig 6.1 shows a estrogen receptor. The estrogen receptor is a protein with several functional parts: a DNA-binding domain, two activation domains, and an estrogen-binding domain.

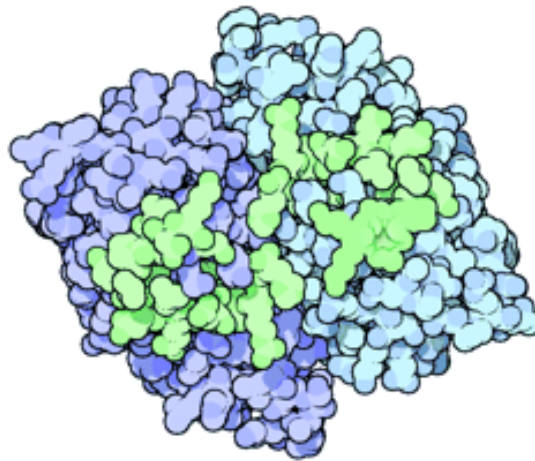


Fig 6.1

(e) It is known that estrogen receptors are nuclear receptors. Suggest their role in the cell. [2]

- **estrogen binds to estrogen receptor in the cytoplasm;**
- **forming hormone-receptor complex;**
- **complex act as a transcriptional factor;**
- **Diffuse into nucleus;**
- **Bind to promoter/ enhancer of proto-oncogene;**
- **Activates proto-oncogene;**
- **Protein controlling normal cell growth are synthesized; leading to normal cell growth**

(f) Which step of multi-step process of cancer is Bevacizumab targeted at? Explain how it leads to the slowing of cancerous growth. [2]

- **Angiogenesis;**
- **Formation of new blood vessels;**
- **Required to bring oxygen and nutrients to the tumor;**
- **By inhibiting angiogenesis, cancer cells will be starved of oxygen and nutrients;**

(g) What is one possible disadvantage of using Bevacizumab? [1]

- **Angiogenesis inhibitors might be toxic to/might affect healthy/normal cells;;**

[Total: 12]

- 7 There are between 325-340 species of hummingbirds found in the Americas. Traditionally, hummingbirds share the same order as the swifts. However, some scientists believe that the hummingbirds evolved relatively recently.

(a) Briefly discuss the process of speciation. [4]

- One or more species arise / split from previously existing species;;
- It involves (any one of the following events):
- Geographical isolation; a population becomes separated from the rest of its species by geographical barrier;;
- Physiological isolation; though reproductively mature at the same time, physiological incompatibility of the gametes make fertilization impossible;;
- Behavioural isolation; the elaborate courtship or mating routine of one subspecies fail to elicit the response in a potential partner of another subspecies;;
- The isolated population will diverge genetically / have accumulated with different mutations;;
- Will be unable to interbreed and produce fertile offspring with the rest of the species;;

The ruby-throated hummingbird, *Archilochus colubris* (Fig 7.1), is found along Mississippi River in North America.

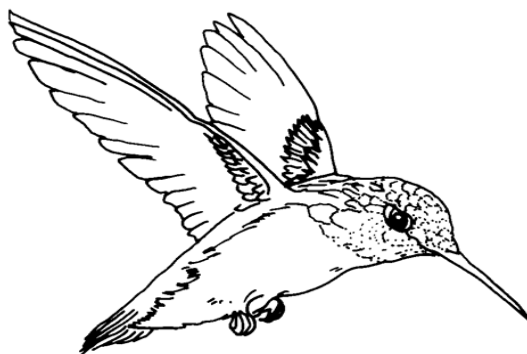


Fig 7.1

(b) Complete Table 7.1 to show the classification of the ruby-throated hummingbird. [4]

Table 7.1

kingdom	<u>Animalia</u>
<u>phylum</u>	Chordata
<u>class</u>	Aves
<u>order</u>	Apodiformes
family	Trochilidae
genus	<u>Archilochus</u>
<u>species</u>	<i>Archilochus colubris</i>

The ruby-throated hummingbirds have long narrow beaks that adapt them to collecting nectar from flowers. Nectar provides energy in the form of sugar, but has very low concentrations of amino acids and protein. Therefore, hummingbirds have to supplement their diet with flying insects. They do this by flying with their beaks wide open towards insects. Scientists have used high-speed video to study hummingbirds' success rate for catching insects. Table 7.2 shows the results, according to which part of the beak the insect touches first.

Table 7.2

Base of beak		Middle of beak		Tip of beak	
Insect successfully caught and swallowed	Insect lost	Insect successfully caught and swallowed	Insect lost	Insect successfully caught and swallowed	Insect lost
65%	35%	44%	56%	17%	83%

[Adapted from source: Yanega, 2004, *Nature*, **428**, p. 615.]

Birds that feed only on flying insects have beaks of different shape from ruby-throated hummingbirds.

(c) State the relationship between the part of the beak that insects first touch and the success rate for catching insects. [1]

- **The nearer to the base the insect touches the higher the success rate for catching insects;;**

(d) Suggest a reason for the relationship in (c). [1]

- **Difficult to manipulate insect at tip of long narrow beak;**
 - **Beak wider open near tip;**
 - **Base of beak is nearer the esophagus / throat / mouth;**
 - **More difficult to escape if insect near base / easier to hold near base;**
- Do not award mark if answer states that the “base of beak is wider than tip”.**

(e) Data showed that there is a higher proportion of hummingbirds with short wide beaks. Account for the predominance of these hummingbirds through natural selection. [2]

- **At selective advantage;**
- **Higher success rate / greater chance for catching insects / lesser chance of losing insects;**
- **Survive to reproduce;**
- **Pass on favorable/advantageous traits/genes/alleles to future generation;**

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

8 (a) Describe the control mechanisms involved in the expression of the eukaryotic genes.

[12]

Features	Eukaryotic Gene Expression
Chromatin Modification;	• DNA methylation prevents gene expression when methyl groups are added to the DNA/CpG / DNA demethylation enhance gene expression when methyl groups are removed from the DNA; • DNA methyltransferase/ DNA demethylase; • Histone acetylation increases gene expression by adding acetyl groups to histones/Histone deacetylation decreases gene expression; • histone acetyltransferase/ histone deacetylase;
Regulatory sequences;	• Basal/general transcription factors; • eg TFIID binds to TATA box; • and RNA polymerase; • bind to proximal control elements/core promoter; • which are required for transcription of all genes; • Specific transcription factors /activators ; • bind to distal control elements / enhancer; (which act in a tissue-specific / time-specific manner) • activate transcription of these specific genes; • repressor; • bind to silencer; • inactivate transcription of these genes;

[max 4]

Post transcription;	• 5' end of the new mRNA is modified by addition of a 7-methyl guanosine cap; • 5' cap helps to protect mRNA from degradation by hydrolytic enzymes/acts as signal for ribosomes to attach/facilitates the transport of mature mRNA out of the nucleus; • pre-mRNA contains a polyadenylation sequence near the 3' end; • a poly-A tail (consisting of 50-250 adenine nucleotides) added to the 3' end; • The length of poly-A tail determines the stability of mRNA/The longer the tail, the more stable is the mRNA/facilitates the transport of matured mRNA out of the nucleus; • RNA splicing; • Spliceosome; • cuts at specific points to release the intron/excise the introns; • joins together/splice the two exons together; • Alternative RNA splicing; • Different mRNA molecules are produced from the same primary transcript; [max 4]
Translation;	• Blocking the initiation stage of polypeptide synthesis; • Regulatory proteins bind to specific sequences on the 5' end of mRNA and block the binding of ribosomes; • Regulating the protein factors (e.g initiation factors) required for initiating translation; • Chemical modification/ Phosphorylation and dephosphorylation of initiation factors activate and repress translation in eukaryotes respectively; • Controlling the lifespan of mRNA molecules enables control over the number of protein molecules that may be translated from it;
Post Translation;	• Chemical modification / adding or removing functional groups, reversibly alters the activity of an enzyme; • Polypeptides must be transported to targeted destinations in the cell for their proper functioning; • Protein can be marked for degradation; • by attaching ubiquitin to them; • Ubiquitin is then recognized by a big protein complex called proteasome (for degradation);

(b) With reference to the three different stages of cell signaling, describe in detail the sequence of events when an insulin molecule reaches the liver cell. [8]

- **Signal reception;**
- **insulin binds specifically to a tyrosine-kinase receptor (RTK);**
- **on the plasma membrane of the target cell / liver cell;**
- **insulin is complementary in shape to receptor site/specific site on the receptor;**
- **recognize and bind to the receptor;**
- **conformational change in the receptor;**
- **dimerization of tyrosine-kinase receptor;**

- **Signal transduction;**
- **the subunit of one insulin receptor phosphorylates the other / the two polypeptides phosphorylate each other / tyrosine kinase parts of the 2 polypeptides add phosphates to the tyrosine tail of the other polypeptide;**
- **binds/phosphorylated and activate relay proteins/insulin response proteins;**
- **relay proteins initiate a signal transduction pathway / insulin response proteins pass signal down by binding to additional proteins;**

- **Cellular response;**
- **activation of glycogen synthase;**
- **Increases the rate of conversion of glucose into glycogen in liver/muscle cells;;**
- **glycogenesis;**

- **Increases in the rate of cell respiration using glucose as respiratory substrate / to breakdown glucose into CO₂ and water;;**
- **Increases the conversion of glucose to fat in adipose cells;;**
- **Hence bringing down the blood glucose level back to normal;**

[Max 8]

9 (a) Distinguish between DNA replication and transcription.

[5]

	DNA Replication	Transcription
1	Occurs before mitosis/meiosis during interphase)	Occurs before translation (during protein synthesis)
2	Uses both strands of DNA as templates;	Uses only one of the strands as template;
3	The elongation of DNA occurs in a bidirectional manner on both templates;	The elongation of mRNA occurs in 5'→3' direction on single template;
4	Original DNA double helix separated, semi-conservative;	No separation of original DNA double helix;
5	Entire DNA molecule unwinds and undergoes replication	Only portion of the sense strand of DNA unwinds and is transcribed
6	Requires DNA polymerase ;	Requires RNA polymerase;
7	Deoxyribonucleotides used	Ribonucleotides used
8	Base-pairing between nucleotides occurs in the form of A=T and C≡G;	Base-pairing between nucleotides occurs in the form of A=U and C≡G;
9	Begins at regions of DNA known as origins of replication;	Begins at regions of DNA known as promoter;
10	Requires RNA priming;	Does not require RNA priming;
11	Product does not undergo post-transcriptional modifications;	Product undergoes post-transcriptional modifications;
12	Produces 2 DNA molecules/synthesize a chain of deoxyribonucleotides;	Produces 1 RNA molecules/synthesize a chain of ribonucleotides;

(b) Describe briefly how enzymes are synthesized within a eukaryotic cell. [15]

1. **Enzymes are protein in nature;**
 2. **Transcription - synthesis of an mRNA molecule with a base sequence complementary to a section of DNA;**
 3. **RNA polymerase attach to the promoter;**
 4. **DNA double helix unwinds;**
 5. **One DNA strand acts as a template for the formation of mRNA;**
 6. **Free nucleotides or ribonucleoside triphosphates are used;**
 7. **According to complementary base-pairing rule: A-U, T-A, G-C, C-G;**
 8. **nucleotides are linked via phosphodiester bond;**
 9. **growing RNA elongates in the 5' to 3' direction;**
 10. **RNA polymerase reach terminator;**
 11. **Polymerase molecule detaches from the RNA molecule and the gene;**
-
12. **Pre-mRNA transcript undergoes post-transcriptional modification;**
 13. **5' end of the new mRNA is modified by addition of a 7-methyl guanosine cap;**
 14. **5' cap helps to protect mRNA from degradation by hydrolytic enzymes/acts as signal for ribosomes to attach/facilitates the transport of mature mRNA out of the nucleus;**
 15. **pre-mRNA contains a polyadenylation sequence/AAUAAA near the 3' end;**
 16. **a poly-A tail (consisting of 50-250 adenine nucleotides) added to the 3' end;**
 17. **The length of poly-A tail determines the stability of mRNA/The longer the tail, the more stable is the mRNA/facilitates the transport of matured mRNA out of the nucleus;**
 18. **Introns are removed;**
 19. **Exons are spliced together;**

20. mRNA leaves the nucleus and enters the cytoplasm;

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21. Translation - the mechanism by which the sequence of bases in mRNA is converted into a sequence of amino acids in a polypeptide chain;
 22. tRNA molecules carry specific amino acid at 3' end, depending on its specific anti-codon;
 23. forming amino-acyl-tRNA complex;
 24. using enzyme amino acyl-tRNA synthetase;
 25. mRNA binds to a small ribosomal subunit;
 26. Aminoacyl-tRNA complex with anticodon UAC/initiator tRNA will bind to the start codon/AUG at the P site;
 27. binding of large ribosomal subunit;
 28. formation of translation initiation complex;
 29. two codons exposed in ribosomes;
 30. A second aminoacyl-tRNA with anticodon complementary to next codon will be brought into A site;
 31. peptide bond forms between the amino acid on the tRNA in the A site with the amino acid/polypeptide bound to tRNA in P site;
 32. catalysed by peptidyl transferase;
 33. ribosome moves on three bases/ a codon/ a triplet downstream;
 34. tRNA in P site moves to E site and released;
 35. tRNA in the A site is translocated to the P site;
 36. elongation repeated until ribosome reaches a stop codon/UAA/UGA/UAG;
 37. release factor binds directly to the stop codon and causes addition of a water molecule;
 38. hydrolyses the polypeptide from the tRNA and release the polypeptide from the ribosome;
 39. translational complex are disassembled;
 40. polypeptide chain folds into its specific shape due to hydrophobic interactions, disulphide, ionic and hydrogen bonds between amino acids;
 41. an enzyme with an active site for binding to substrate is formed;

[Max 15]